

HOSPITAL MANAGEMENT SYSTEM

Python Programming Project Report

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INTRODUCTION TO PROBLEM SOLVING AND PROGRAMMING

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1. Introduction

The Hospital Management System is a menu-driven Python application designed to demonstrate basic patient and appointment management. It allows a user to register patients, view patient records, search for a patient, delete a patient record, book an appointment, and view appointments.

2. Problem Statement

Managing patient details manually can be time-consuming and may make it difficult to find or update information. This project demonstrates how a simple computer program can organize patient details and appointment information through a menu-based interface.

3. Objectives

- Create a simple menu-driven application using Python.
- Register patients using a unique patient ID.
- Display and search registered patient details.
- Delete a patient record when requested.
- Book appointments for registered patients.
- Display appointment information.
- Practise functions, dictionaries, lists, loops, conditionals, and user input.

4. Software and Requirements

- Programming language: Python 3
- Code editor: Visual Studio Code (VS Code)
- Interface: Terminal / command line
- Data structures: Dictionary for patients and list for appointments
- Optional text file: patients.txt (file saving must be verified separately)

5. System Features

1. **Register Patient:** Collects a patient ID, name, age, and disease, then adds the record to the program's patient dictionary.
2. **View Patients:** Displays the registered patient records when the view function is implemented and connected correctly.
3. **Search Patient:** Searches for a patient using the entered patient ID.
4. **Delete Patient:** Removes a patient record using its ID; the current program may also remove associated appointments.
5. **View Appointments:** Displays appointments already booked.
6. **Book Appointment:** Checks that the patient ID exists, then collects doctor, date, and time details.
7. **Exit:** Ends the menu loop and closes the application.

6. Program Logic / Algorithm

1. Start the program.
2. Create the patient dictionary and appointment list.
3. Display the main menu.
4. Read the user's choice.
5. Call the corresponding function for registration, viewing, searching, deletion, appointment booking, or appointment viewing.
6. For appointment booking, check whether the entered patient ID exists.
7. Display the result of the selected operation.
8. Repeat the menu until the user selects Exit.
9. Stop the program.

7. Sample Test Cases

Test	Action	Expected result
1	Register a new patient	Success message and patient record added in memory
2	View registered patients	Registered patient details displayed
3	Search using a valid ID	Matching patient details displayed

4	Delete using a valid ID	Patient removed and confirmation displayed
5	View appointments when none exist	Message indicating no appointments
6	Book appointment for registered ID	Appointment details collected and saved in memory
7	Book appointment for unknown ID	Prompt to register the patient first
8	Enter an invalid menu choice	Invalid-choice message displayed

8. Limitations

- In the current in-memory version, patient and appointment data are lost when the program exits.
- Text-file persistence should only be claimed after saving and reloading records has been tested.
- The application is an educational prototype and is not intended for real clinical use.
- Authentication, privacy controls, input validation, and robust date/time validation can be added in future versions.

9. Future Enhancements

- Save and reload patient and appointment records using a text file or SQLite database.
- Improve validation for IDs, ages, names, and appointment dates.
- Add appointment cancellation and rescheduling.
- Add a simple graphical user interface.
- Add login and access controls if required by the project specification.

10. Conclusion

The Hospital Management System demonstrates how Python functions, conditional statements, loops, dictionaries, and lists can be combined to create a basic management application. Testing each menu option helps verify that the functions work as expected. Persistent storage and additional validation are useful next steps before treating the project as complete.

11. Screenshots

The following screenshots were taken from the VS Code terminal and show the main menu and each tested menu option of the program.

```
===== HOSPITAL MANAGEMENT SYSTEM =====  
1. Register Patient  
2. View Patients  
3. Search Patient  
4. Delete Patient  
5. View Appointments  
6. Book Appointment  
7. Exit
```

Figure 1: Main menu of the Hospital Management System.

```
Enter your choice: 1  
Enter Patient Registration ID: P1001  
Enter patient name: Adya  
Enter patient age: 18  
Enter disease: Typhoid  
Patient registered successfully!
```

Figure 2: Successful registration of a new patient (ID P1001).

```
Enter your choice: 2
```

```
--- Registered Patients ---
```

```
Patient ID: P1001
```

```
Name: Adya
```

```
Age: 18
```

```
Disease: Typhoid
```

Figure 3: Viewing the registered patients.

```
Enter your choice: 3
```

```
Enter Patient ID to search: P1001
```

```
Patient found!
```

Figure 4: Searching for a patient using a valid Patient ID.

```
Enter your choice: 4
```

```
Enter Patient ID to delete: P1003
```

```
No patient found with that ID.
```

Figure 5: Delete option entered with an ID that is not registered (P1003), showing the "No patient found" message.

```
Enter your choice: 6
```

```
Enter registered Patient ID: P1001
```

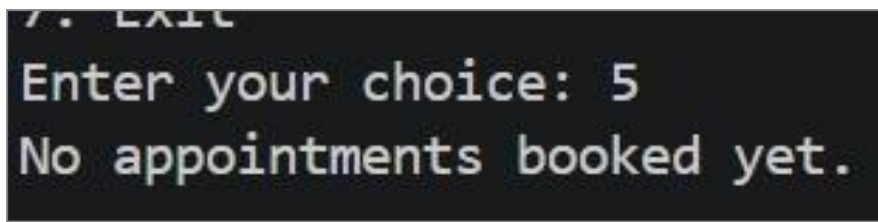
```
Enter doctor's name: Nitin
```

```
Enter appointment date (DD-MM-YYYY): 03-10-2026
```

```
Enter appointment time: 12:00 Pm
```

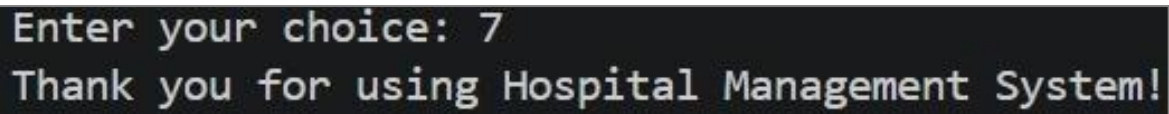
```
Appointment booked successfully!
```

Figure 6: Successful appointment booking for a registered patient.



```
7. EXIT  
Enter your choice: 5  
No appointments booked yet.
```

Figure 7: Viewing appointments when none have been booked yet.



```
Enter your choice: 7  
Thank you for using Hospital Management System!
```

Figure 8: Exiting the program.

Declaration: This report should be reviewed and edited to match the exact code, features, and requirements of the submitted project.